

PROGRAM:

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import random

def choose_word(word_list):
    # Choose a random word and convert it to uppercase
    return random.choice(word_list).upper()

def display_board(missed_letters, correct_letters, secret_word):
    print("\nMissed Letters:", " ".join(missed_letters))

    display = ""

    for letter in secret_word:
        if letter in correct_letters:
            display += letter + " "
        else:
            display += "_ "

    print("Word:", display)

def get_guess(already_guessed):
    while True:
        guess = input("Guess a letter: ").upper()

        if len(guess) != 1:
            print("Please enter a single letter.")
        elif not guess.isalpha():
            print("Please enter a LETTER.")
        elif guess in already_guessed:
            print("You have already guessed that letter.")
        else:
            return guess

def check_game_over(correct_letters, missed_letters, secret_word):
    won = True

    for letter in secret_word:
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        if letter not in correct_letters:
            won = False
            break
    if won:
        print("\nCongratulations! You guessed the word:", secret_word)
        return True
    if len(missed_letters) >= 7:
        print("\nGame Over! You ran out of guesses.")
        print("The word was:", secret_word)
        return True
    return False

def play_again():
    return input("\nDo you want to play again? (yes or no): ").lower().startswith('y')

def play():
    word_list = [
        "python", "jupyter", "programming", "function",
        "Analytics", "Explore", "Generative",
        "Infinite", "Network", "Foundation"
    ]
    secret_word = choose_word(word_list)
    missed_letters = []
    correct_letters = []
    while True:
        display_board(missed_letters, correct_letters, secret_word)
        guess = get_guess(missed_letters + correct_letters)
        # Update guessed letters
        if guess in secret_word:
            correct_letters.append(guess)

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        print("Correct Guess!")
    else:
        missed_letters.append(guess)
        print("Wrong Guess!")
    game_over = check_game_over(
        correct_letters,
        missed_letters,
        secret_word
    )
    if game_over:
        if play_again():
            missed_letters = []
            correct_letters = []
            secret_word = choose_word(word_list)
        else:
            break
# Start the game
play()
```